AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# M110
# Python Programming

## Midterm Examination (MTA) - Makeup
## Summer Semester 2025/2026

**Time Allowed: 2 hours**

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**Instructions:**

- Total Marks: **30**
- This exam consists of **Two** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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## PART 1: Multiple Choices questions [5 Marks]

**Question1:** Answer the following questions by choosing the most correct statement.

1- How do you access the last element of a list named **my_list** in Python?
   a. my_list[0]
   b. my_list[len(my_list)]
   c. my_list.last()
   d. my_list[-1]

2- Which of the following is **required** in Python if statements?
   a. Curly braces
   b. Indentation
   c. Semicolons
   d. Parentheses

3- Based on what you have learned in this course so far, which loop is used for situations where you don't know the needed number of iterations in advance?
   a. Do-while loop
   b. For loop
   c. while loop
   d. Both choices b and c

4- Which statement skips the **current iteration** and continues with the next?
   a. return
   b. continue
   c. stop
   d. break

5- Which operator checks whether an item exists in a list?
   a. ==
   b. is in
   c. in
   d. Either b or c

## PART 2: Problem Solving Questions [25 Marks]

**Question 2:** [10 marks]

What is the output of the following program?

**a- 2.5 marks**

```
x = 12
if x % 2 == 0:
    print(x)
    x-=3
print("Done")
```

**b- 2.5 marks**

```
for i in range(1,6,2):
    print(i * 2, end=" ")

print ("\nDone!")
```

**c- 2.5 marks**

```
n = 7
while n >= 0:
    if n % 2 == 0:
        print(n, end=" ")
    n = n - 1
print("End")
```

**d- 2.5 marks**

```
total = 0
for i in range(1, 4):
    for j in range(i):
        total = total + j
print(total)
```

**Question 3:** [4 marks]

**Create a flowchart for the following problem:**

You are given a grade score (from 0 to 100).

- If the score is 50 or above, print "Pass".
- If the score is below 50, print "Fail".

Design a flowchart that represents this logic clearly using standard flowchart symbols.

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**Question 4**: [5 marks]
Given a list of student exam scores, write a Python program that calculates the average score, finds the highest score achieved, and prints the results.
**scores = [78, 85, 92, 64, 89, 75, 81, 95, 70, 88]**
*Expected output:*
*Average Score: 81.7*
*Highest Score: 95*

**Question 5**: [6 marks]
You are given a tuple of integers. Write a Python program to create a new list. This new list should contain the same elements in the given tuple, but with all odd numbers doubled, and display at the end the original Tuple and the new list sorted in ascending order.
In your code, you should use **one loop** and **one if statement**.
**input_tuple = (5, 8, 11, 14, 17)**
*Expected output:*
*Original Tuple: (5, 8, 11, 14, 17)*
*New List: [8, 10, 14, 22, 34]*

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**End of questions**

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